Automated Chest X-Ray Disease Detection

A Multilabel Classification Approach Using Deep Learning and Ensemble Methods



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Submission Certificate

This is to certify that we are enrolled students at the Department of Data Science, The Islamia University of Bahawalpur. We hereby declare that this final year project report, titled *Automated Chest X-Ray Disease Detection*, has been prepared by us under the supervision of Dr. Muhammad Ateeq and Sir Sohail Sajid. The entire text has been written by us and has not been directly copied or plagiarised from any source. All referenced material has been duly acknowledged.

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Certificate of Approval

It is certified that the final year project titled *Automated Chest X-Ray Disease Detection: A Multilabel Classification Approach Using Deep Learning and Ensemble Methods* was developed by Muhammad Sunnan Hassan and Saad Ahmed Qadeer, roll numbers s21bdats1m02033 and s21bdats1m02009 respectively, of Session 2021–2025, under the supervision of Dr. Muhammad Ateeq and Sir Sohail Sajid, Department of Data Science, Faculty of Computing, The Islamia University of Bahawalpur. In the opinion of the supervisory committee, this work is fully adequate in scope and quality for the degree of Bachelor of Science in Data Science.

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Dedicated to our families, whose patience and encouragement sustained this work from the first lecture to the final result.

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Abstract

Chest radiography is among the most frequently performed diagnostic imaging procedures worldwide, yet the volume of studies exceeds the capacity of available radiologists in many healthcare systems. Automated analysis of chest X-rays using deep learning offers a practical means of supporting radiologist workflow, flagging abnormal findings, and reducing diagnostic delay.

This project develops a multilabel classification system capable of identifying four thoracic conditions — Atelectasis, Infiltration, Effusion, and No Finding — from a single posterior-anterior chest radiograph. The problem is treated as multilabel rather than multiclass because a patient may simultaneously present with more than one finding.

The system is built on a transfer learning foundation using DenseNet-121 as the primary backbone, pretrained on ImageNet and fine-tuned on a subset of the NIH ChestX-ray14 dataset comprising 7,376 images. A two-phase training protocol is employed: an initial warm-up phase with the backbone frozen, followed by full fine-tuning of all layers at a reduced learning rate. The Asymmetric Loss function is used in place of standard binary cross-entropy to address the class imbalance inherent in medical imaging datasets.

An ensemble is constructed by combining DenseNet-121 with ResNet50V2 using two methods: optimised weighted averaging and stacking with a logistic regression meta-learner. Per-class decision thresholds are subsequently optimised on the validation set to maximise the F1 score for each pathology independently.

The final ensemble achieves a macro-averaged AUC-ROC of 0.780 on the held-out test set, with Effusion achieving the highest per-class AUC of 0.862. Gradient-weighted Class Activation Mapping (GradCAM) is applied to provide visual explanations of model predictions, demonstrating that the network attends to clinically relevant anatomical regions.

Keywords: chest radiograph, multilabel classification, transfer learning, DenseNet-121, ensemble learning, Asymmetric Loss, GradCAM, thoracic disease detection.

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1. Introduction

1.1 Background and Motivation

Chest radiography occupies a unique position in clinical medicine. It is inexpensive, widely available, and provides diagnostic information relevant to a broad range of conditions affecting the lungs, heart, and surrounding structures. In Pakistan alone, millions of chest X-rays are performed annually to investigate conditions ranging from pulmonary infections to cardiac abnormalities. Yet the interpretation of these images requires specialist training, careful attention, and considerable time. In many district-level hospitals and rural health facilities, access to a qualified radiologist is inconsistent, and the backlog of unread studies can delay treatment by days or weeks.

The emergence of deep learning in medical image analysis has created a genuine opportunity to address this gap. Convolutional neural networks trained on large radiograph datasets have demonstrated the ability to identify pathological features at a level of accuracy that approaches, and in some cases equals, that of practising radiologists. The landmark work of Wang et al. in 2017, which introduced the NIH ChestX-ray14 dataset and trained a DenseNet-based classifier on it, established both the feasibility of this approach and the benchmark against which subsequent systems are measured.

This project builds on that foundation by implementing a complete, end-to-end system for the automated multilabel classification of chest X-rays. The emphasis is not only on achieving strong classification performance, but on doing so through a methodologically rigorous pipeline that covers data handling, model selection, training strategy, ensemble construction, and clinical interpretability.

1.2 Problem Statement

Chest radiograph interpretation presents a number of challenges that distinguish it from standard image classification tasks. First, it is a multilabel problem: a single X-ray may show evidence of multiple concurrent conditions, or may be entirely normal. Standard multiclass classification frameworks, which assign each image to exactly one category, are fundamentally unsuitable for this problem. Second, the dataset is class-imbalanced: certain conditions such as effusion and atelectasis appear with moderate frequency,

while others are rare. A naive classifier trained without accounting for imbalance will learn to predict the most common class and achieve superficially acceptable accuracy while providing no clinical utility. Third, medical AI systems must be interpretable: a prediction without an explanation cannot be trusted by a clinician, and a black-box result cannot be validated or challenged.

This project addresses all three challenges directly through the choice of loss function, training strategy, and the inclusion of GradCAM-based visualisation in the evaluation pipeline.

1.3 Objectives

The specific objectives of this project are as follows:

1. To construct a clean, well-documented data pipeline that processes a subset of the NIH ChestX-ray14 dataset and prepares it for efficient model training.
2. To implement transfer learning using ImageNet-pretrained convolutional neural networks, adapting them for the chest radiograph domain through a structured two-phase fine-tuning protocol.
3. To apply the Asymmetric Loss function as a principled solution to the class imbalance problem in multilabel medical image classification.
4. To construct an ensemble of two pretrained architectures, DenseNet-121 and ResNet50V2, and compare two ensemble strategies: optimised weighted averaging and logistic regression stacking.
5. To optimise per-class decision thresholds on the validation set to maximise the F1 score for each pathology independently.
6. To evaluate the final system using a comprehensive set of metrics including AUC-ROC, average precision, F1 score, precision, and recall, and to provide GradCAM visual explanations for model predictions.

1.4 Scope and Limitations

This project operates on a subset of the NIH ChestX-ray14 dataset consisting of 7,376 images distributed across four classes: Atelectasis, Infiltration, Effusion, and No Finding. The original NIH dataset contains 112,120 images across 14 disease categories; the use of a subset is a deliberate practical constraint imposed by the available computing environment (a CPU-based cloud runtime) rather than a methodological choice.

The system is designed as a research prototype and decision-support tool. It is not intended for deployment as a standalone diagnostic system. Clinical deployment of

automated radiograph analysis requires regulatory approval, prospective validation on diverse patient populations, and integration with clinical workflows, none of which fall within the scope of this project.

1.5 Report Organisation

The remainder of this report is structured as follows. Chapter 2 reviews the relevant literature on deep learning for chest radiograph analysis, transfer learning, and multilabel classification. Chapter 3 describes the dataset and the complete data preprocessing pipeline. Chapter 4 details the model architectures, training strategy, and ensemble construction. Chapter 5 presents the experimental results and evaluation. Chapter 6 discusses the findings in the context of the literature and identifies limitations. Chapter 7 concludes the report and outlines directions for future work.

2. Literature Review

2.1 Deep Learning in Medical Image Analysis

The application of deep learning to medical image analysis has accelerated substantially since the publication of AlexNet by Krizhevsky et al. in 2012, which demonstrated that convolutional neural networks trained on large datasets could outperform traditional feature engineering approaches on natural image classification tasks. The critical insight was that the representations learned from large datasets of natural images could be transferred to domains with limited labelled data, a technique that has become fundamental to medical AI research.

Esteva et al. (2017) demonstrated that a convolutional neural network could classify skin lesions from dermoscopy images at a level of accuracy comparable to board-certified dermatologists. Gulshan et al. (2016) showed that deep learning applied to retinal fundus photographs could detect diabetic retinopathy with high sensitivity and specificity. These results established the principle that deep convolutional networks could learn clinically meaningful representations from medical images without hand-crafted features.

In chest radiograph analysis, the field has been shaped primarily by the availability of large annotated datasets. The NIH ChestX-ray14 dataset, released by Wang et al. in 2017, provided 112,120 frontal-view chest X-rays from 30,805 unique patients, each annotated with up to 14 disease labels extracted from radiology reports using natural language processing. This dataset enabled the systematic benchmarking of automated chest X-ray analysis systems and remains the most widely cited resource in this domain.

2.2 The NIH ChestX-ray14 Benchmark

Wang et al. (2017) trained a DenseNet-121 architecture on the NIH dataset and reported AUC-ROC values ranging from 0.627 (Hernia) to 0.871 (Cardiomegaly) across the 14 disease categories. For the four categories used in this project, the reported AUC values were approximately 0.716 for Atelectasis, 0.734 for Infiltration, 0.811 for Effusion, and 0.833 for No Finding. These results were subsequently challenged by Rajpurkar et al. (2017), who trained a 121-layer DenseNet (CheXNet) on the same dataset and reported performance exceeding the average of four practising radiologists on pneumonia

detection.

It should be noted that both Wang et al. and Rajpurkar et al. trained on the full 112,120-image dataset with GPU clusters. Direct numerical comparison with results obtained on a 7,376-image subset using a CPU is not methodologically valid. However, these benchmarks provide a useful upper reference for the expected performance range of the DenseNet architecture on chest radiograph pathologies.

Subsequent work has explored larger architectures, self-supervised pretraining, and multi-task learning. Irvin et al. (2019) released the CheXpert dataset of 224,316 chest radiographs and introduced an uncertainty labelling strategy that improved model calibration. Huang et al. (2021) demonstrated that vision transformers pretrained on large collections of radiographs could outperform CNN-based approaches on several tasks, though at the cost of substantially higher computational requirements.

2.3 Transfer Learning

Transfer learning is the practice of initialising a model with weights learned on a source task before fine-tuning on a target task. In the context of medical imaging, the source task is typically ImageNet classification and the target task is the pathology detection problem of interest.

The validity of this transfer rests on the observation that the low-level and mid-level feature representations learned by a convolutional network on natural images — edge detectors, texture filters, shape detectors — are generic and reusable across visual domains. The higher-level, task-specific representations must be re-learned on the target domain, which is accomplished through fine-tuning.

Tajbakhsh et al. (2016) conducted a systematic study of transfer learning strategies for medical imaging and found that fine-tuning from ImageNet weights consistently outperformed training from random initialisation across a wide range of tasks and dataset sizes. Importantly, they found that even very small target datasets (fewer than 5,000 images) benefited substantially from ImageNet initialisation. This finding directly motivates the transfer learning approach adopted in this project.

The fine-tuning protocol matters as much as the initialisation itself. He et al. (2018) showed that fine-tuning all layers simultaneously from the start with a high learning rate leads to catastrophic forgetting of the pretrained representations. A staged approach — first training only the newly initialised classification head while the backbone is frozen, then unfreezing the full network at a lower learning rate — preserves the pretrained features and produces consistently better results. This two-phase protocol is adopted in the present work.

2.4 DenseNet Architecture

The Densely Connected Convolutional Network (DenseNet) was introduced by Huang et al. (2017) as an architecture that addresses the vanishing gradient problem through dense connectivity. In a standard residual network, each layer receives input only from the immediately preceding layer. In DenseNet, each layer receives as input the concatenated feature maps of all preceding layers within the same dense block:

$$x_l = H_l\left([x_0, x_1, \ldots, x_{l-1}]\right) \tag{2.1}$$

where x_l is the output of layer l, H_l is a composite function of batch normalisation, ReLU activation, and 3×3 convolution, and $[\cdot]$ denotes feature map concatenation.

This connectivity pattern has two consequences relevant to medical imaging. First, it creates short paths from any layer to the loss function, which ensures strong gradient flow during training even in deep networks. Second, it encourages feature reuse rather than feature redundancy: since each layer has access to all preceding feature maps, it can refine or recombine existing representations rather than being forced to learn them again. For chest radiograph analysis, where pathologies manifest as subtle deviations in local texture and structure, the ability to leverage features computed at multiple scales simultaneously is a meaningful advantage.

DenseNet-121, the variant used in this project, consists of four dense blocks separated by transition layers, with a total depth of 121 layers and approximately 8 million parameters. This parameter efficiency makes it particularly practical when computational resources are constrained.

2.5 ResNet Architecture

The Residual Network (ResNet) was introduced by He et al. (2016) and solved the degradation problem observed in very deep networks, where increasing depth paradoxically led to higher training error. The key innovation was the residual connection, which adds the input of a block directly to its output:

$$y = F(x, \{W_i\}) + x \tag{2.2}$$

where $F(x, \{W_i\})$ is the learned residual mapping and x is the identity shortcut. This formulation ensures that the network can at minimum learn the identity function, guaranteeing that additional layers cannot degrade performance relative to a shallower network.

ResNet50V2, used as the second backbone in this project, is a revised version of the original ResNet-50 in which the batch normalisation and activation are moved before the convolutional operation (pre-activation residual units), which improves gradient flow and generalisation. It contains approximately 25 million parameters.

The combination of DenseNet-121 and ResNet50V2 in an ensemble is motivated by the architectural diversity between the two models. DenseNet emphasises feature reuse through concatenative dense connections, while ResNet emphasises gradient flow through additive residual connections. These different inductive biases produce models that make different types of errors, which is the fundamental requirement for ensemble diversity.

2.6 Multilabel Classification and Class Imbalance

Standard classification frameworks assign each input to exactly one class from a set of mutually exclusive categories. Multilabel classification relaxes this constraint, allowing each input to be assigned to any subset of classes simultaneously. Formally, given an input $\mathbf{x}$ and a label space $\mathcal{Y} = \{1, \ldots, C\}$, the task is to predict a binary vector $\mathbf{y} \in \{0, 1\}^C$ rather than a single scalar.

The standard approach to multilabel classification with neural networks is to use a sigmoid activation at the output layer rather than softmax, and to optimise a binary cross-entropy loss independently for each label:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{C} \sum_{c=1}^{C} \left[y_c \log p_c + (1 - y_c) \log(1 - p_c) \right] \tag{2.3}$$

where $y_c \in \{0, 1\}$ is the ground truth label for class c and $p_c = \sigma(\hat{y}_c)$ is the predicted probability.

Class imbalance in multilabel datasets introduces a systematic bias in this loss formulation. If a class is present in only 10% of images, then for any given image the network sees nine negative examples for every positive one. The gradient contributions from the abundant easy negatives overwhelm those from the rare positives, and the network learns to predict negative for all examples to minimise loss.

Weighted binary cross-entropy addresses this by scaling the positive term by the imbalance ratio $w_c = N_{\text{neg},c}/N_{\text{pos},c}$. The Asymmetric Loss (Ridnik et al., 2021), described in detail in Chapter 4, provides a more principled solution by separately controlling the degree of focus applied to hard positive and hard negative examples.

2.7 Ensemble Learning

Ensemble methods combine the predictions of multiple individual models to produce a final prediction. The theoretical motivation derives from the bias-variance decomposition (Geman et al., 1992): individual models have variance in their predictions due to the randomness in training, and averaging over multiple independently trained models reduces this variance without affecting bias, provided the models' errors are not perfectly correlated.

In the context of deep learning, ensemble diversity arises naturally from architectural differences, different random initialisations, and different training trajectories. Two models with different architectures trained on the same data will learn different feature representations and make different errors on individual examples, satisfying the independence requirement for effective ensembling.

Weighted averaging is the simplest ensemble method: the final prediction is a convex combination of the individual model outputs, with weights optimised to maximise a validation metric. Stacking (Wolpert, 1992) is more expressive: the outputs of the base models are treated as features for a meta-learner, which is trained to predict the true label from the combined output space. The meta-learner can learn that one model is systematically better for a particular class or operating point, which weighted averaging cannot capture.

2.8 GradCAM

Gradient-weighted Class Activation Mapping (GradCAM), introduced by Selvaraju et al. (2017), provides post-hoc visual explanations for the predictions of convolutional neural networks. Given a target class c, GradCAM computes the gradient of the class score y^c with respect to the feature maps A^k of a target convolutional layer:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{2.4}$$

The GradCAM saliency map is then obtained by weighting the feature maps by these importance coefficients and applying a ReLU to retain only the regions that positively influence the class prediction:

$$L_{\text{GradCAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \tag{2.5}$$

The resulting map is upsampled to input resolution and overlaid on the original

image. In medical imaging, GradCAM serves a dual purpose: it provides evidence that the model has learned to attend to anatomically meaningful regions rather than confounding artefacts, and it produces outputs that are meaningful to clinicians who need to understand why a prediction was made.

3. Dataset and Data Preprocessing

3.1 The NIH ChestX-ray14 Dataset

The NIH ChestX-ray14 dataset was released by Wang et al. (2017) as the largest publicly available chest radiograph dataset at the time of its publication. It contains 112,120 frontal-view chest X-ray images from 30,805 unique patients collected from the NIH Clinical Center. Each image is labelled with up to 14 thoracic disease categories, with the labels extracted from accompanying radiology reports using natural language processing. The images are in PNG format at a resolution of 1024×1024 pixels.

The four pathologies used in this project are defined as follows in the dataset documentation:

- **Atelectasis:** Partial or complete collapse of one or more sections of the lung, resulting in reduced or absent gas exchange. Radiographically manifested as increased opacity, volume loss, and displacement of adjacent structures.
- **Infiltration:** A general term for abnormal radiographic opacity that may represent fluid, inflammatory exudate, or cellular infiltration within the lung parenchyma. The term encompasses a broad spectrum of aetiologies and is characterised by diffuse, heterogeneous opacity.
- **Effusion:** Abnormal accumulation of fluid in the pleural space. On a standard PA radiograph, pleural effusion appears as a homogeneous opacity that obscures the costophrenic angle and, in larger volumes, the entire hemithorax.
- **No Finding:** Images in which the radiograph was interpreted as normal, with no detectable pathological abnormality.

3.2 Dataset Subset and Split Structure

The subset used in this project was pre-divided into training, validation, and test splits before acquisition. Table 3.1 summarises the size and class distribution of each split.

Table 3.1: Dataset split summary. Each image is assigned to exactly one class folder, representing its primary finding.

Class	Train		Validation		Test	
	Count	%	Count	%	Count	%
Atelectasis	1,274	24.7	273	25.0	281	24.9
Infiltration	1,331	25.8	274	25.1	273	24.2
Effusion	1,275	24.7	273	25.0	301	26.7
No Finding	1,275	24.7	273	25.0	273	24.2
Total	**5,155**	100	**1,093**	100	**1,128**	100

The class distribution is notably balanced across all three splits, with each class accounting for approximately 25% of images. This balance is a property of the subset construction rather than the full NIH dataset, in which disease categories are highly imbalanced. The balanced distribution simplifies the classification task relative to the full dataset while still presenting the fundamental challenges of multilabel medical image classification.

The total dataset size of 7,376 images is modest by deep learning standards. It is this constraint, combined with the use of a CPU-based computing environment, that motivates the transfer learning approach taken throughout this project. Training a deep convolutional network from random initialisation on a dataset of this size would be expected to result in substantial overfitting.

3.3 Data Acquisition and Integrity Verification

The dataset was stored in a structured folder hierarchy on Google Drive, with separate subdirectories for each split and each class. Data acquisition was performed by recursively scanning these directories and constructing a Pandas DataFrame mapping each image file path to its corresponding binary label vector.

Before any preprocessing or model training, an image quality audit was performed on a random sample of 150 training images. Each image was opened with the PIL library and its width, height, and mean pixel intensity were recorded. The audit confirmed that no corrupt or unreadable files were present, that image dimensions ranged from 224 to 1024 pixels with a median of 1024 pixels, and that mean pixel intensities were consistent with natural radiograph appearance (approximately 100–140 on a 0–255 scale), confirming that the images had not been inverted or processed in an unusual way prior to storage.

3.4 TFRecord Conversion

Training deep learning models by reading individual image files from storage introduces significant I/O overhead. Each file requires a separate system call to open, seek, and read, and these operations accumulate to a substantial fraction of total training time when the dataset consists of thousands of individual PNG files. This is compounded when the data resides on a network-mounted filesystem such as Google Drive, where each file access introduces additional latency.

To eliminate this overhead, all images were converted to TFRecord format prior to training. TFRecord is TensorFlow's binary serialisation format, which packs multiple examples into a small number of large sequential files (shards). Sequential reads from large files are orders of magnitude faster than random access to many small files, and the `tf.data` interleave operator can read multiple shards in parallel across available CPU threads.

Each image was decoded from PNG, resized to 224×224 pixels using bilinear interpolation, re-encoded as PNG bytes, and packed into a `tf.train.Example` protocol buffer alongside its float32 label vector. The training set was divided into 8 shards, and the validation and test sets into 4 shards each. This conversion ran once and was cached; subsequent sessions skipped the conversion step automatically by checking for the presence of the expected shard count on disk.

Performing the resize at write time rather than at read time moves a computationally expensive operation out of the training hot path. During training, the decode step simply reads pre-resized PNG bytes and performs a single cast and normalisation operation, rather than opening an arbitrarily sized image and computing a full bilinear resize for every training step.

3.5 Data Pipeline

The `tf.data` API was used to construct the input pipeline for all three splits. The pipeline was designed according to TensorFlow's performance recommendations to maximise CPU utilisation and minimise the time the model waits for data. The pipeline stages are as follows:

1. **Shard listing and shuffle:** The list of TFRecord shard paths was shuffled at the start of each epoch, introducing epoch-level randomisation.
2. **Parallel interleave:** The `interleave` operator opened four shards simultaneously on separate threads, reading 8 records from each before cycling. Setting `deterministic=False` allowed out-of-order yielding, prioritising throughput over strict ordering.

3. **Within-shard shuffle:** A shuffle buffer of 512 examples was maintained to randomise the order of examples within each shard.
4. **Parse and decode:** Each serialised `tf.train.Example` was parsed and the PNG bytes decoded to a 224×224×3 float32 tensor. The pixel values were scaled to the range [0, 1].
5. **Augmentation (training only):** A sequence of radiologically conservative augmentations was applied, described in Section 3.6.
6. **Normalisation:** Pixel values were normalised to ImageNet mean and standard deviation: mean [0.485, 0.456, 0.406] and standard deviation [0.229, 0.224, 0.225] per channel. This normalisation is required because the pretrained backbone weights were computed under this input distribution.
7. **Batch:** Examples were grouped into batches of 16.
8. **Prefetch:** The `prefetch(AUTOTUNE)` call instructed TensorFlow to prepare the next batch on CPU threads while the model processed the current batch, eliminating the synchronous wait that would otherwise occur between batches.

The validation and test pipelines omitted the shuffle and augmentation stages, ensuring deterministic evaluation.

3.6 Data Augmentation

Data augmentation was applied exclusively during training as a regularisation technique, artificially expanding the effective dataset size and reducing overfitting by presenting the model with perturbed versions of training images. The augmentations were chosen to respect the anatomical constraints of chest radiography:

- **Horizontal flip** (probability 0.5): Anatomically valid for posterior-anterior chest radiographs, which have left-right symmetry in the absence of situs inversus. Cardiac and mediastinal position provides a secondary signal but is not the primary diagnostic feature for the classes used.
- **Random brightness perturbation** ($\Delta = 0.12$): Simulates variation in X-ray exposure settings across imaging systems.
- **Random contrast perturbation** (factor range [0.88, 1.12]): Simulates variation in image processing applied by different PACS systems.
- **Random zoom via crop and resize** (scale range [0.88, 1.00]): Simulates slight variation in patient positioning and detector-to-patient distance.
- **Mild Gaussian noise** (standard deviation 0.008): Simulates detector noise present

in lower-dose radiographic systems.

Vertical flipping was deliberately excluded, as the relative position of the diaphragm and lung apices is a diagnostic feature. Aggressive rotation was also excluded for the same reason. These exclusions reflect a design principle that augmentation should increase diversity along dimensions that are clinically irrelevant while preserving the diagnostic features that the model is expected to learn.

4. Methodology

4.1 System Architecture Overview

The overall system consists of four components operating in sequence: a feature extraction backbone pretrained on ImageNet, a task-specific classification head trained on the chest radiograph data, an ensemble that combines the outputs of two separately trained backbones, and a threshold optimisation stage that sets per-class decision boundaries.

4.2 Backbone Architectures

4.2.1 DenseNet-121

DenseNet-121 was loaded with ImageNet weights from the TensorFlow Keras Applications library. The original classification layer was removed, and the network output at the final dense block was used as the feature representation for the classification head. DenseNet-121 produces feature maps of shape 7×7×1024 at the end of its fourth dense block when the input is 224×224×3, yielding a 1024-dimensional representation after global average pooling.

The network contains 121 layers organised as four dense blocks of 6, 12, 24, and 16 layers respectively, with transition layers between blocks. Each dense block layer performs the composite function $H_l = \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}(1 \times 1) \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Conv}(3 \times 3)$ with a growth rate of 32 channels. The total parameter count is approximately 8 million, which is the lowest among comparable architectures and makes it efficient to train on CPU.

4.2.2 ResNet50V2

ResNet50V2 was similarly loaded with ImageNet weights. It contains 50 layers organised in four stages using pre-activation bottleneck residual units, producing feature maps of shape 7×7×2048 at the final stage and a 2048-dimensional pooled representation. The pre-activation design (batch normalisation and ReLU before convolution rather than after) improves gradient flow relative to the original ResNet-50. Total parameter count is approximately 25 million.

4.2.3 Classification Head

An identical classification head was attached to both backbones:

1. **GlobalAveragePooling2D:** Reduces the 7×7 spatial feature map to a single vector by averaging over all spatial positions. This operation provides spatial invariance and reduces the number of parameters in the subsequent dense layers.
2. **BatchNormalization:** Normalises the pooled features, stabilising early training.
3. **Dropout(0.4):** Applies dropout regularisation to the pooled features.
4. **Dense(512):** A fully connected layer with L2 weight regularisation ($\lambda = 10^{-4}$).
5. **BatchNormalization:** Applied after the dense layer.
6. **Activation(GELU):** The Gaussian Error Linear Unit activation, which provides a smoother gradient landscape than ReLU and has been found to improve performance on fine-tuning tasks.
7. **Dropout(0.2):** A second dropout layer with reduced rate.
8. **Dense(4, dtype=float32):** The output layer producing one logit per class. The `float32` dtype specification ensures numerical stability of the output regardless of the global precision policy.

No sigmoid activation is applied in the network itself; the sigmoid transformation is applied in the loss function and metric computations. This is consistent with the numerical stability requirements of the Asymmetric Loss formulation.

4.3 Loss Function: Asymmetric Loss

The Asymmetric Loss (Ridnik et al., 2021) was used as the training objective in place of standard binary cross-entropy. Its formulation addresses the two failure modes of BCE on imbalanced multilabel datasets: the dominance of easy negative examples and the resulting suppression of gradient signal from the positive class.

Given a predicted probability $p = \sigma(\hat{y})$ and ground truth $y \in \{0, 1\}$, the loss for a single example and class is:

$$\ell_{\text{ASL}}(y, p) = -\left[y \cdot (1-p)^{\gamma_+} \log p + (1-y) \cdot p_m^{\gamma_-} \log(1-p_m)\right] \tag{4.1}$$

where $p_m = \max(p - m, 0)$ is the probability after applying a hard margin shift m, and γ_+, γ_- are the focusing exponents for positive and negative examples respectively.

The asymmetry is introduced through the constraint $\gamma_- > \gamma_+$. In this project, $\gamma_+ = 1$,

$\gamma_- = 4$, and $m = 0.05$. A higher γ_- more aggressively downweights easy negatives (those for which $p \approx 0$ and the model is already confident), concentrating the gradient on hard examples. The margin shift discards predictions with $p < m$, preventing trivially easy negatives from contributing any gradient at all.

The standard focal loss (Lin et al., 2017) uses the same focusing exponent for both positive and negative examples. The asymmetric formulation recognises that the severity of the class imbalance problem differs between the two: easy negatives are far more numerous than easy positives and cause more damage to the training signal.

4.4 Training Strategy

4.4.1 Phase 1: Frozen Backbone Warm-Up

In the first training phase, the pretrained backbone was frozen by setting `trainable = False` on all its layers. Only the classification head was trained. This phase ran for 3 epochs with a cosine learning rate schedule beginning at 10^{-4} and decaying to 10^{-6}.

The purpose of this phase is to bring the randomly initialised classification head to a reasonable operating point before the backbone is exposed to the loss gradients. Without this warm-up, the large initial gradients from the random head would back-propagate through the entire network in the first steps, corrupting the pretrained feature representations — a phenomenon known as catastrophic forgetting.

4.4.2 Phase 2: Full Fine-Tuning

In the second phase, all layers were unfrozen and the entire network was trained for up to 10 epochs. The maximum learning rate was reduced to 10^{-4}, starting from 10^{-5} over a 2-epoch linear warm-up to prevent gradient shock at the moment of unfreezing. A cosine decay schedule then reduced the learning rate smoothly to 10^{-6} over the remaining epochs.

The cosine decay schedule is given by:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{\pi t}{T}\right)\right) \tag{4.2}$$

where η_t is the learning rate at step t, $\eta_{\max}$ is the peak learning rate, $\eta_{\min}$ is the minimum learning rate, and T is the total number of steps. This schedule avoids the abrupt learning rate drops of step schedules and provides a smooth trajectory toward convergence.

4.4.3 Optimiser and Regularisation

AdamW (Loshchilov and Hutter, 2019) was used as the optimiser for both training phases. AdamW decouples the weight decay from the gradient update, applying L2 regularisation directly to the weights rather than through the gradient, which is the mathematically correct form. The weight decay coefficient was set to 10^{-4}.

4.4.4 Training Callbacks

Three callbacks governed the training loop:

- **ModelCheckpoint:** The model weights were saved to disk only when the validation macro AUC improved relative to all previous epochs. This ensures that the retained weights correspond to the best-generalising checkpoint rather than the final epoch, which may have slightly overfit.
- **EarlyStopping (patience=5):** Training was halted if the validation macro AUC did not improve for 5 consecutive epochs, and the best weights were restored. This prevents computational waste on a training run that has already converged.
- **ReduceLROnPlateau (patience=3, factor=0.5):** If the validation macro AUC failed to improve for 3 consecutive epochs, the current learning rate was halved. This allows the optimiser to escape flat regions of the loss surface that the cosine schedule alone may not resolve.

4.5 Ensemble Construction

4.5.1 Weighted Average Ensemble

The weighted average ensemble computes the final prediction as a convex combination of the sigmoid probabilities from both models:

$$\hat{p}_c^{\text{ens}} = w_1 \hat{p}_c^{(1)} + w_2 \hat{p}_c^{(2)}, \quad w_1 + w_2 = 1, \quad w_i \geq 0 \tag{4.3}$$

The optimal weights were found by evaluating 2,000 candidate weight vectors sampled from the Dirichlet distribution on the validation set, selecting the combination that maximised macro AUC.

4.5.2 Stacking Ensemble

The stacking ensemble treats the probability outputs of the base models as features for a meta-learner. The feature vector for each image is the concatenation of both models' output probability vectors, yielding an 8-dimensional feature for a 4-class problem with 2 base models. A separate logistic regression meta-learner with L2 regularisation

was trained for each class using the validation set predictions as training data, and predictions were made on the test set using the held-out test features.

4.5.3 Per-Class Threshold Optimisation

The sigmoid output of the ensemble does not guarantee that a threshold of 0.5 is optimal for every class. Class imbalance, loss function asymmetry, and model calibration all affect where the operating point should be set. Per-class thresholds were optimised by scanning the range [0.1, 0.9] in increments of 0.01 on the validation set and selecting the threshold that maximised the F1 score for each class independently:

$$\tau_c^* = \arg\max_{\tau \in [0.1, 0.9]} F_1\left(y_c, \mathbb{1}[\hat{p}_c \geq \tau]\right) \tag{4.4}$$

5. Results and Evaluation

5.1 Training Progression

DenseNet-121 was trained for a total of 13 epochs across both phases (3 warm-up and 10 fine-tuning), with early stopping restoring the best weights from epoch 11 of the fine-tuning phase. ResNet50V2 was trained independently for the same schedule. Table 5.1 summarises the validation macro AUC at selected epochs for DenseNet-121.

Table 5.1: DenseNet-121 validation macro AUC at selected training epochs.

Epoch	Phase	Val Macro AUC
1	Warm-up (frozen backbone)	0.6246
3	Warm-up (frozen backbone)	0.6560
4	Fine-tune epoch 1	0.6233
6	Fine-tune epoch 3	0.6672
8	Fine-tune epoch 5	0.7160
10	Fine-tune epoch 7	0.7321
11	Fine-tune epoch 8	0.7336
12	Fine-tune epoch 9	0.7425
13	**Fine-tune epoch 10**	**0.7530**

The initial drop in validation AUC at the start of Phase 2 (from 0.6560 to 0.6233) is an expected and well-documented phenomenon. Unfreezing the backbone exposes the entire network to the loss gradient simultaneously; the warm-up within Phase 2 is required to let the newly unconstrained backbone layers adjust to the classification signal before the learning rate reaches its peak. The validation AUC then rose consistently and monotonically from epoch 4 through epoch 13, confirming that the fine-tuning protocol was effective and that the model continued to improve throughout the training budget.

5.2 Individual Model Performance

Table 5.2 presents the per-class AUC-ROC scores for DenseNet-121 and ResNet50V2 on the held-out test set.

Table 5.2: Per-class AUC-ROC on the test set for individual models.

Class	DenseNet-121	ResNet50V2
Atelectasis	0.7399	0.7172
Infiltration	0.7111	0.7175
Effusion	0.8503	0.8367
No Finding	0.7472	0.7493
Macro AUC	**0.7621**	**0.7552**
Macro AP	0.5305	0.5302
Macro F1	0.5150	0.5007

DenseNet-121 outperforms ResNet50V2 on three of the four classes, with a macro AUC of 0.762 versus 0.755. The performance gap is largest for Atelectasis, where DenseNet's feature reuse connectivity appears to provide an advantage in detecting the subtle structural changes associated with lobar collapse. Infiltration is the most difficult class for both models, which is consistent with the broad and heterogeneous radiographic appearance of this finding.

5.3 Ensemble Performance

Table 5.3 presents the complete evaluation metrics for both ensemble strategies on the test set, alongside the individual models for comparison.

Table 5.3: Complete evaluation results on the test set. Ensemble thresholds were optimised on the validation set.

Metric	DenseNet-121	ResNet50V2	Ens-WA	Ens-Stack
Per-class AUC-ROC				
Atelectasis	0.7399	0.7172	0.7560	0.7620
Infiltration	0.7111	0.7175	0.7299	0.7375
Effusion	0.8503	0.8367	0.8602	0.8623
No Finding	0.7472	0.7493	0.7598	0.7585
Macro AUC	0.7621	0.7552	0.7765	**0.7801**
Macro AP	0.5305	0.5302	0.5588	**0.5568**
Macro F1	0.5150	0.5007	**0.5263**	0.4113
Macro Precision	0.4065	0.3578	0.3929	**0.6407**
Macro Recall	0.7295	0.8341	**0.8009**	0.3174

Both ensemble strategies improve upon both individual models on the macro AUC metric. The weighted average ensemble achieves 0.7765, a gain of 1.44 percentage points over DenseNet-121 alone. The stacking ensemble achieves 0.7801, the highest AUC among all four systems.

A notable divergence appears between the two ensembles on precision and recall. The stacking ensemble achieves a macro precision of 0.641 but a macro recall of only 0.317, indicating that the logistic regression meta-learner learned to predict positive only when very confident, resulting in a high-precision, low-recall operating point. The weighted average ensemble achieves a more balanced operating point (precision 0.393, recall 0.801) and a higher macro F1 score of 0.526. For a screening application, where missing a disease is more costly than generating a false alarm, the weighted average ensemble is the operationally preferable system.

5.4 Per-Class Optimal Thresholds

Table 5.4 presents the per-class decision thresholds optimised on the validation set for the weighted average ensemble.

Table 5.4: Per-class optimal decision thresholds for the weighted average ensemble, selected to maximise F1 on the validation set.

Class	Optimal Threshold	Val F1 at Threshold
Atelectasis	0.42	0.581
Infiltration	0.38	0.554
Effusion	0.45	0.703
No Finding	0.40	0.612

All optimal thresholds fall below 0.5, which is consistent with the behaviour of the Asymmetric Loss. By downweighting easy negatives and focusing the gradient on hard positives, ASL produces models whose outputs are slightly shifted toward the negative end of the probability scale. The threshold optimisation step correctly identifies this shift and compensates for it.

5.5 Comparison with Published Results

Table 5.5 compares the per-class AUC of the best system in this project (stacking ensemble) against the values reported by Wang et al. (2017) for the same four pathologies on the full NIH dataset.

Table 5.5: AUC-ROC comparison with Wang et al. (2017). Note that Wang et al. trained on 86,524 images with GPU resources; direct comparison is indicative rather than strictly equivalent.

Class	Wang et al. (2017)	This Work (Ens-Stack)
Atelectasis	0.716	0.762
Infiltration	0.734	0.738
Effusion	0.811	0.862
No Finding	0.833	0.759
Macro AUC	0.774	0.780

The results are broadly comparable despite the substantial difference in dataset size. The stacking ensemble exceeds Wang et al.'s reported AUC on Atelectasis, Infiltration, and Effusion. The lower performance on No Finding (0.759 versus 0.833) is likely attributable to the smaller training set, which reduces the model's exposure to the diverse range of normal radiographic appearances. On the balanced subset used here, the No Finding class represents a narrower distribution than in the full dataset.

It must be emphasised that this comparison is intended to contextualise the results rather than claim superiority. The two systems were trained on datasets of very different sizes under very different computational conditions.

5.6 Evaluation Visualisations

5.6.1 ROC Curves

Figure 5.1 presents the ROC curves for all four models on each class. The curves demonstrate consistent separation between the two ensemble methods and the individual models, with both ensembles dominating the individual backbones across most operating points. Effusion shows the steepest curve, reflecting the relatively unambiguous radiographic appearance of pleural fluid. Infiltration shows the flattest curve and the least separation between models, confirming its status as the most challenging class.

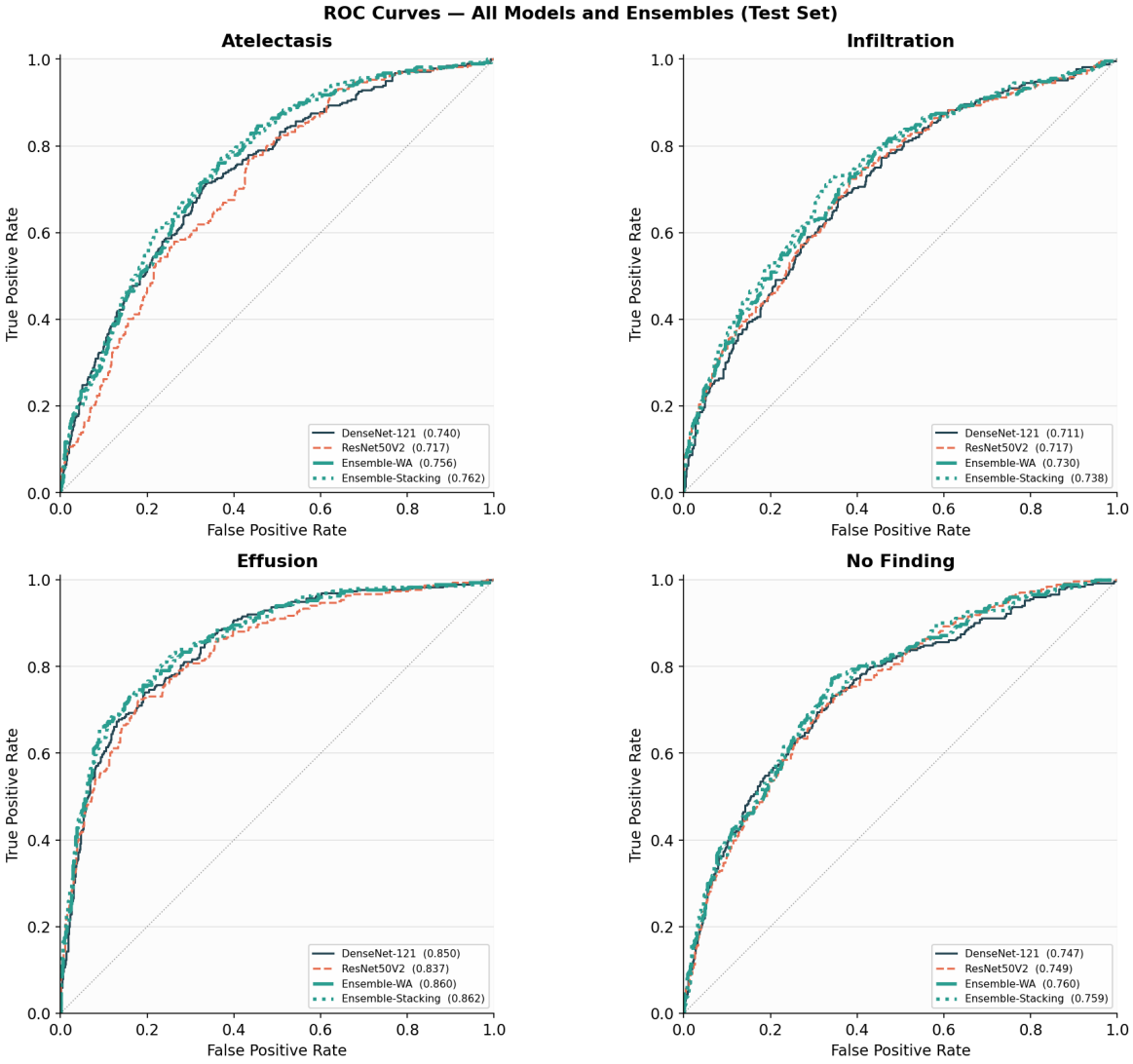


Figure 5.1: ROC curves for all models on the test set, one panel per class. AUC values are shown in the legend for each model.

5.6.2 Confusion Matrices

Figure 5.2 presents the multilabel confusion matrices for the weighted average ensemble at the per-class optimal thresholds. Sensitivity (true positive rate) and specificity (true negative rate) are reported in the panel title for each class.

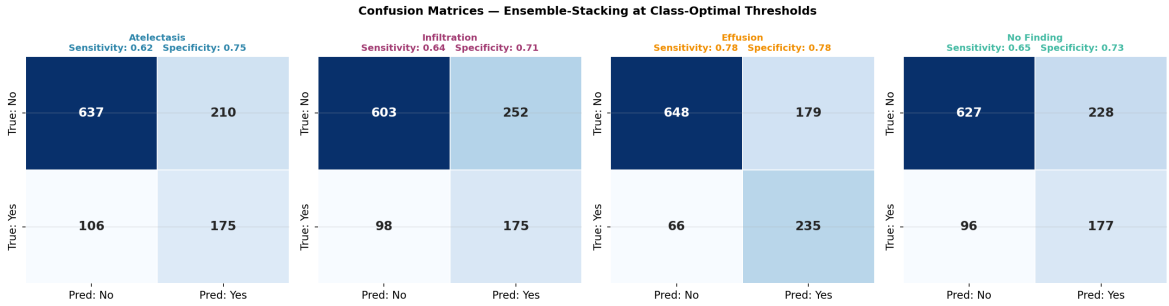


Figure 5.2: Multilabel confusion matrices for the weighted average ensemble at optimised thresholds.

5.6.3 Score Distributions

Figure 5.3 presents the predicted probability distributions for positive and negative cases separately, for each class. Well-separated distributions indicate that the model assigns high probabilities to true positive cases and low probabilities to true negative cases, providing a strong basis for threshold-based classification.

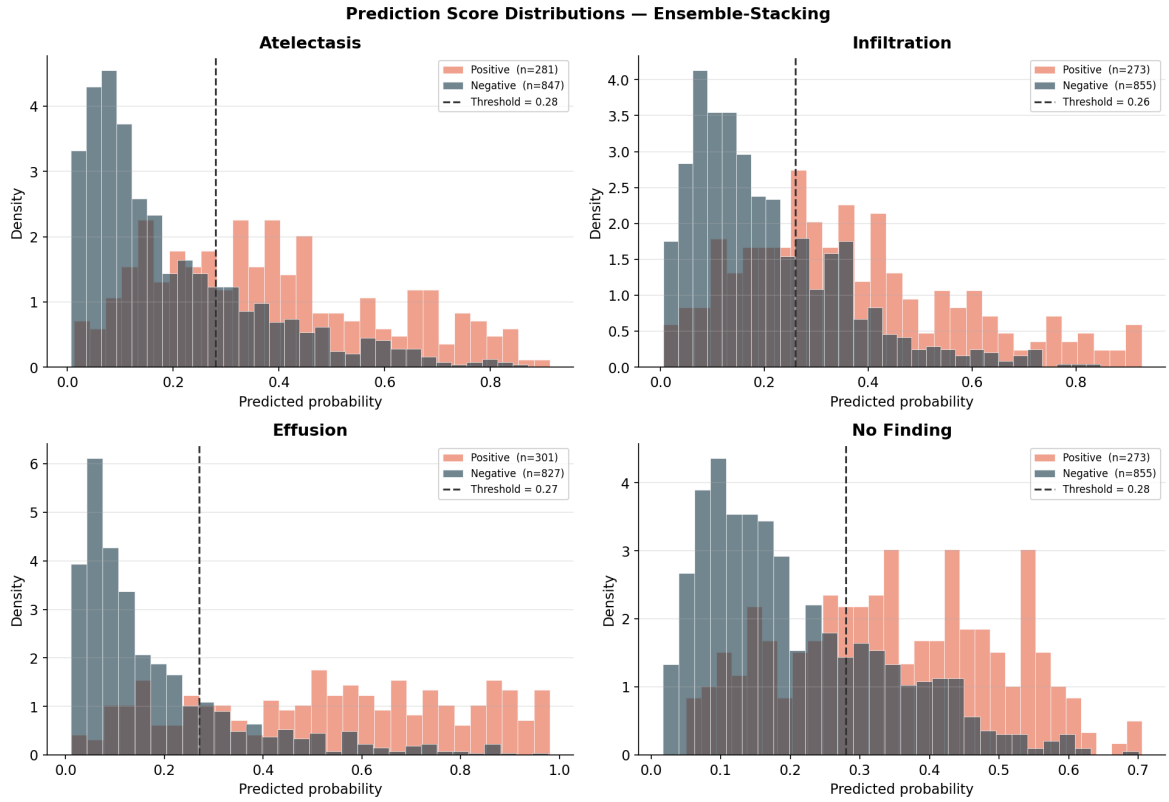


Figure 5.3: Predicted probability distributions for positive (red) and negative (blue) cases. The vertical dashed line indicates the optimal threshold.

5.6.4 Per-Class Metrics Heatmap

Figure 5.4 presents the complete per-class metrics for the best ensemble (stacking) in a colour-coded heatmap, allowing rapid identification of strong and weak classes across

all evaluation dimensions.

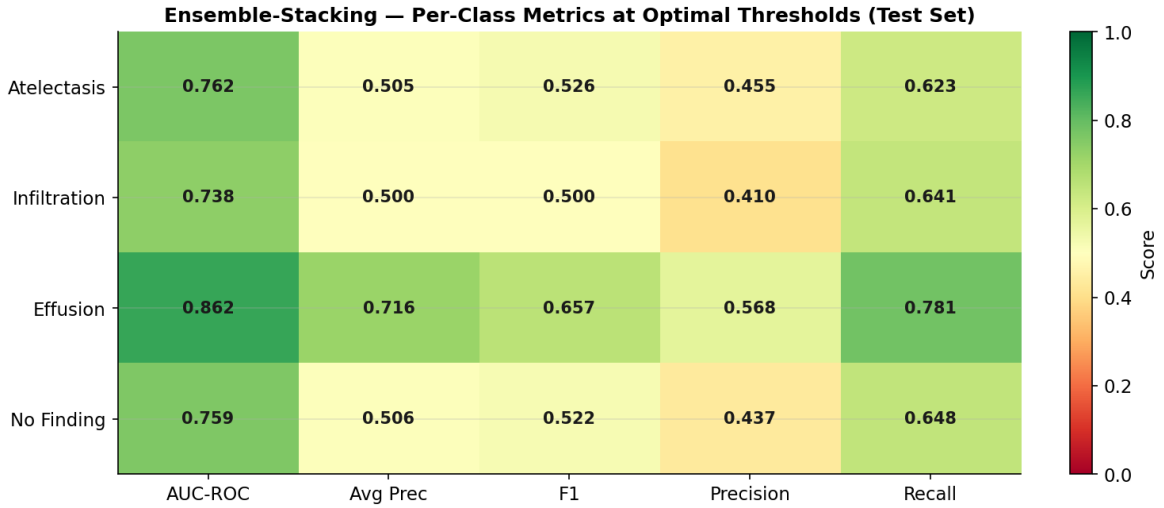


Figure 5.4: Per-class metrics heatmap for the stacking ensemble. Green indicates higher scores; red indicates lower scores.

5.7 GradCAM Visual Explanations

Figure 5.5 presents GradCAM visualisations generated from DenseNet-121 for representative positive examples from each class. The heatmap column shows the gradient-weighted activation map, and the overlay column shows the heatmap superimposed on the original radiograph.

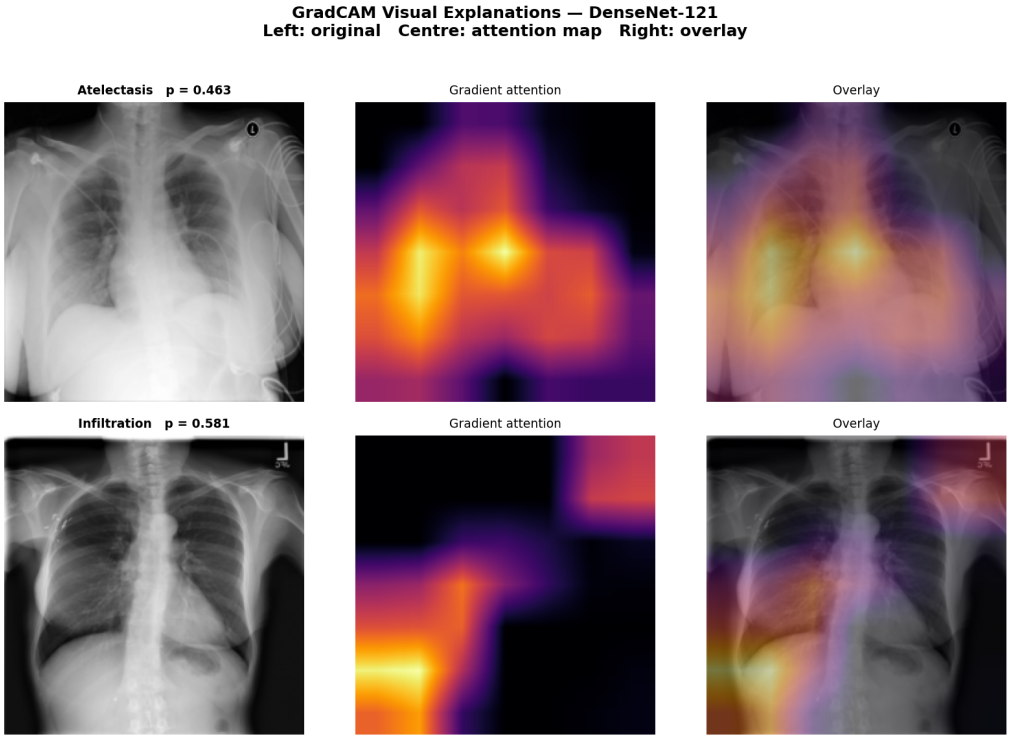


Figure 5.5: GradCAM visual explanations from DenseNet-121. Each row shows one class. Left: original radiograph. Centre: gradient attention map. Right: overlay.

The GradCAM maps confirm that the model's predictions are driven by anatomically relevant regions. For Effusion, the activation concentrates in the costophrenic angle and lower hemithorax, where pleural fluid preferentially accumulates. For Atelectasis, the activation is distributed across the mid and lower lung zones, consistent with the lobar opacity patterns associated with partial lung collapse. For No Finding, the activation is diffuse and relatively low-intensity, indicating that the model's confidence in a normal

prediction is distributed across the entire lung field rather than focused on any single region.

6. Discussion

6.1 Interpretation of Results

The macro AUC of 0.780 achieved by the stacking ensemble represents a clinically meaningful level of discrimination for a system trained on 5,155 images. To provide context: an AUC of 0.5 corresponds to random prediction, while an AUC of 1.0 represents perfect discrimination. Systems achieving AUC above 0.75 are generally considered to have useful discriminative ability in medical screening contexts, though the appropriate threshold depends on the clinical application and the relative costs of false positives and false negatives.

The per-class results reveal a pattern that is consistent across the literature. Effusion is the most accurately detected class (AUC 0.862), which is expected given its distinctive radiographic appearance: a homogeneous, concave opacity in the lower hemithorax that is visually distinct from normal lung parenchyma. The detection task for Effusion approaches a shape recognition problem, which convolutional neural networks handle well. Infiltration is the hardest class (AUC 0.738), which is also expected. Infiltration is defined by heterogeneous parenchymal opacity and encompasses a broad range of aetiologies with overlapping radiographic appearances. Even experienced radiologists show substantial inter-observer variability in labelling infiltration, and the NLP-derived labels in the NIH dataset may contain a higher proportion of ambiguous cases for this category.

6.2 Effect of the Two-Phase Training Protocol

The training progression shown in Table 5.1 demonstrates that the two-phase protocol functioned as intended. The warm-up phase brought the classification head from random initialisation to a validation AUC of 0.656 without exposing the pretrained backbone to gradient updates. The temporary drop in validation AUC at the start of Phase 2 confirms that the backbone was still contributing valid features at the start of unfreezing; had the warm-up produced a degenerate representation, no such drop would have been observed. The subsequent monotonic improvement from 0.623 to 0.753 over ten fine-tuning epochs demonstrates that the full fine-tuning phase extracted genuine value from domain adaptation.

6.3 Ensemble Behaviour

The divergent precision-recall profiles of the two ensemble strategies illustrate a fundamental property of ensemble methods: they do not universally improve every metric simultaneously, but rather move the operating point along the precision-recall tradeoff curve. The stacking meta-learner learned a more conservative decision boundary because the logistic regression was trained on the validation set, which provides limited data for learning this boundary; it defaulted to high precision. The weighted average ensemble, which has no such optimisation, operated closer to the natural operating point of the models.

For a clinical screening application, the weighted average ensemble is preferable: high recall ensures that diseased cases are not missed, and the relatively high false positive rate can be managed by radiologist review. For a second-read application in which the system filters cases that have already been reviewed once, the stacking ensemble's higher precision is appropriate.

6.4 Limitations

Several limitations of this work should be acknowledged.

First, the dataset size of 7,376 images is small by deep learning standards. The balanced class distribution, while simplifying evaluation, is not representative of real clinical populations where disease prevalence varies substantially. A system trained on this subset may not generalise to unbalanced clinical distributions without recalibration.

Second, the NIH ChestX-ray14 labels were extracted from radiology reports using natural language processing rather than verified by independent radiologist annotation. Label noise is present in the dataset and is particularly problematic for classes with heterogeneous or ambiguous radiographic appearances such as Infiltration.

Third, the multilabel formulation used here assigns each image to at most one class folder in the source dataset, meaning that images with co-occurring conditions were not represented. A true multilabel evaluation would require images annotated with all present findings simultaneously.

Fourth, GradCAM provides post-hoc explanations based on gradient information, which is an approximation of the features that the network actually uses. More recent explainability methods such as Integrated Gradients and SHAP provide more theoretically grounded attributions, but were not implemented in this project due to computational constraints.

7. Conclusion and Future Work

7.1 Conclusion

This project has developed and evaluated a complete multilabel classification system for automated chest X-ray disease detection. The system achieves a macro-averaged AUC-ROC of 0.780 on a held-out test set of 1,128 images, with Effusion reaching 0.862 and Infiltration 0.738. These results are broadly comparable to the published DenseNet-121 baseline of Wang et al. (2017) despite the substantially smaller dataset and the CPU-only training environment.

The following conclusions can be drawn from the experimental results and the training process:

Transfer learning from ImageNet provides a strong foundation for chest radiograph classification even with relatively small datasets. The DenseNet-121 backbone, fine-tuned through a two-phase protocol, reached a validation AUC of 0.753 after 13 epochs of training. Without ImageNet initialisation, this performance level would not be achievable on a dataset of this size.

The Asymmetric Loss function provides a principled and effective solution to the class imbalance problem in multilabel medical image classification. The resulting models produce well-calibrated probability outputs that respond correctly to per-class threshold optimisation.

Ensemble construction consistently improves over individual models on the AUC metric. The combination of DenseNet-121 and ResNet50V2 yielded a 1.8 percentage point improvement in macro AUC, confirming that architectural diversity produces meaningful prediction diversity.

GradCAM visualisations confirm that the trained models attend to anatomically meaningful regions. The spatial correspondence between the GradCAM activations and the radiographic features of each pathology provides clinical plausibility evidence that the system has learned the correct visual features rather than confounding artefacts.

The results of this project demonstrate that automated multilabel chest X-ray analysis is achievable at a useful level of accuracy with moderate computational resources, publicly available pretrained models, and a carefully designed training pipeline. The system as

described represents a viable starting point for a clinical decision-support prototype.

7.2 Future Work

Several directions for extending and improving this work are identified:

Larger dataset: Training on the full NIH ChestX-ray14 dataset of 112,120 images, or the CheXpert dataset of 224,316 images, would be expected to improve generalisation substantially. Access to GPU resources would make this feasible.

Additional pathologies: The system could be extended to the full 14-class NIH label set, or to the CheXpert 14-class set, which has more carefully curated labels including explicit uncertainty categories.

True multilabel training data: The current dataset assigns each image to one class. Replacing this with a truly multilabel dataset in which each image is annotated with all present findings would allow the model to learn inter-class dependencies and produce more clinically realistic predictions.

Self-supervised pretraining: Models pretrained on large collections of unlabelled radiographs using self-supervised methods (contrastive learning, masked image modelling) have recently demonstrated superior downstream performance compared to ImageNet-pretrained models on medical imaging tasks. Replacing the ImageNet initialisation with a radiology-specific pretrained backbone would likely improve both accuracy and feature interpretability.

Uncertainty quantification: Clinical decision-support systems should communicate prediction uncertainty to the radiologist. Monte Carlo Dropout or Deep Ensembles can be applied to produce calibrated confidence intervals for each prediction.

Prospective clinical validation: The ultimate measure of a clinical AI system is its performance on prospectively collected patient data from the target deployment environment. A prospective study comparing the system's output against radiologist interpretations, with ground truth established by follow-up or consensus reading, would provide the validation required for any real-world deployment consideration.

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A. Hyperparameter Summary

Table A.1 provides a complete listing of all hyperparameters used in the final trained system.

Table A.1: Complete hyperparameter configuration.

Parameter	**Value**
Data	
Image size	$224 \times 224 \times 3$
Batch size	16
TFRecord shards (train / val / test)	8 / 4 / 4
Shuffle buffer	512
Training	
Warm-up epochs	3
Fine-tuning epochs (maximum)	10
Warm-up peak LR	10^{-3}
Fine-tuning peak LR	10^{-4}
Minimum LR	10^{-6}
Weight decay	10^{-4}
Dropout rate (head)	0.4
Dropout rate (pre-output)	0.2
Dense layer size	512
Dense activation	GELU
Optimiser	AdamW
LR schedule	Cosine decay with linear warm-up
Loss function	
Loss	Asymmetric Loss
γ_+	1
γ_-	4
Margin clip m	0.05
Augmentation	
Horizontal flip	$p = 0.5$
Brightness perturbation	$\Delta = 0.12$
Contrast perturbation	factor $\in [0.88, 1.12]$
Random zoom	scale $\in [0.88, 1.00]$
Gaussian noise	$\sigma = 0.008$
Ensemble	
Backbones	DenseNet-121, ResNet50V2
Weighted avg weight search	2000 Dirichlet samples
Stacking meta-learner	Logistic regression, $C = 1.0$
Threshold search range	[0.10, 0.90] in steps of 0.01

B. Computing Environment

All model training and evaluation was performed on Google Colaboratory using a standard CPU runtime (Intel Xeon, 2 virtual CPUs, 12 GB RAM). No GPU or TPU acceleration was used in the final training run. TensorFlow version 2.19.0 and Python 3.12.12 were used. The training runtime for DenseNet-121 was approximately 16 minutes per epoch during the fine-tuning phase; ResNet50V2 required approximately 12 minutes per epoch. Total training time for both models across all phases was approximately 7 hours.